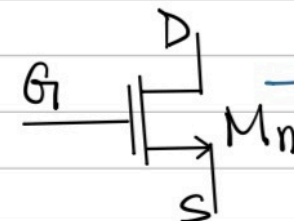


CMOS

→ Complementary MOSFET

→ It uses complementary pairs of NMOS and PMOS to build the logic gates.

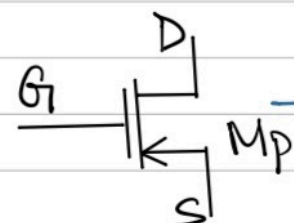
NMOS →  → Threshold Voltage, $V_{tn} > 0$

→ (i) Cutoff : $V_{GS} < V_{tn} \rightarrow I_D = 0$

(on) → $V_{GS} \geq V_{tn}$

→ (ii) Linear / Triode : $V_{DS} < V_{GS} - V_{tn}$
→ $I_D = K_n \{ 2(V_{GS} - V_{tn})V_{DS} - V_{DS}^2 \}$

→ (iii) Saturation : $V_{DS} \geq V_{GS} - V_{tn}$
→ $I_D = K_n (V_{GS} - V_{tn})^2$

PMOS →  → Threshold Voltage, $V_{tp} < 0$

→ (i) Cutoff : $V_{SG} < -V_{tp} \rightarrow I_D = 0$

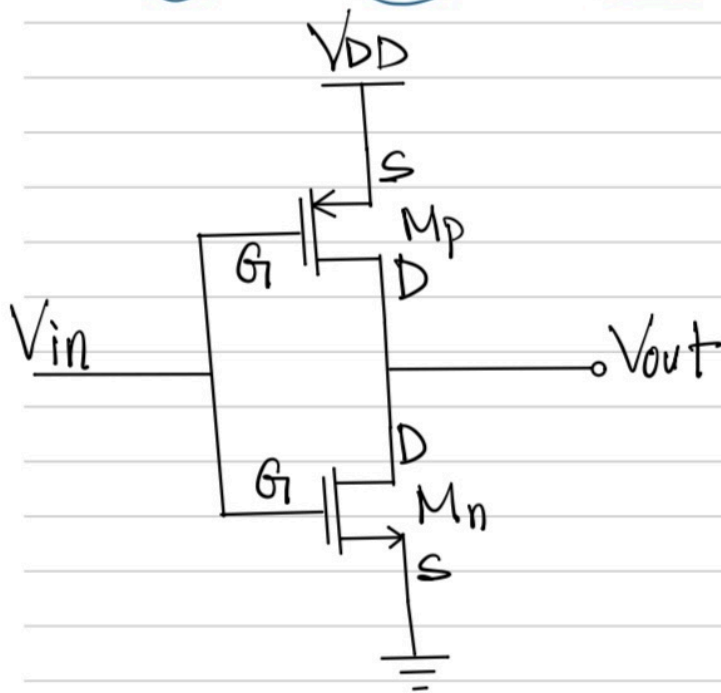
(on) → $V_{SG} \geq -V_{tp}$

→ (ii) Linear / Triode : $V_{SD} < V_{SG} + V_{tp}$
→ $I_D = K_p \{ 2(V_{SG} + V_{tp})V_{SD} - V_{SD}^2 \}$

→ (iii) Saturation : $V_{SD} \geq V_{SG} + V_{tp}$
→ $I_D = K_p (V_{SG} + V_{tp})^2$

⊗ K_n, K_p → Process Parameters of NMOS & PMOS

The CMOS Inverter



for NMOS (M_n)

$$V_G = V_{in}, V_S = 0, V_D = V_{out}$$

$$V_{GS_n} = V_{in} - 0 = V_{in}$$

$$V_{DS_n} = V_{out} - 0 = V_{out}$$

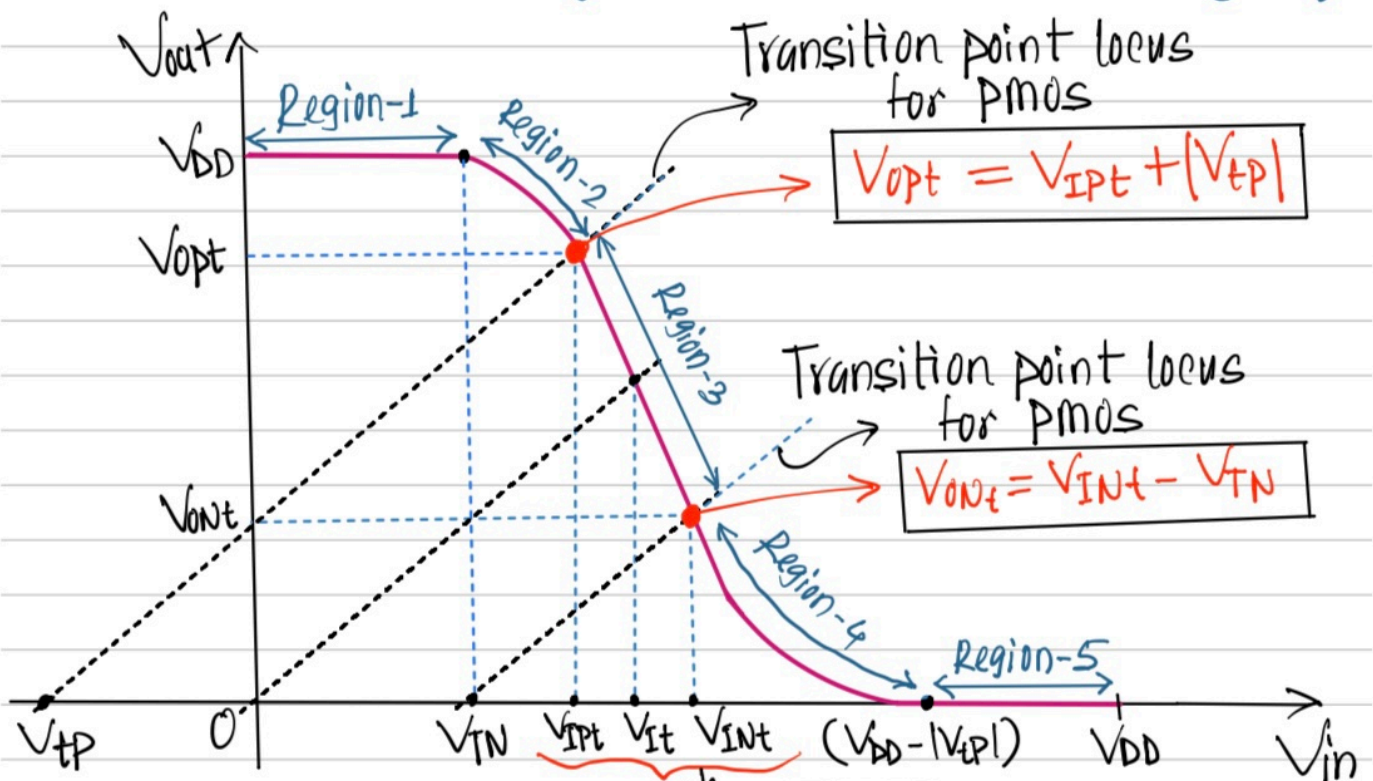
for PMOS (M_p)

$$V_G = V_{in}, V_S = V_{DD}, V_D = V_{out}$$

$$V_{SG_p} = V_{DD} - V_{in}$$

$$V_{SD_p} = V_{DD} - V_{out}$$

CMOS Inverter Voltage Transfer Characteristics (VTC)



* for Ideal VTC $\rightarrow V_{tPT} = V_{tIT} = V_{tINT}$

$\rightarrow V_{tPT}, V_{tPT}$: Transition voltages of PMOS

$\rightarrow V_{tNT}, V_{tNT}$: Transition voltages of NMOS

Mode of operations

- Region-1: $0 \leq V_{in} < V_{TN} \rightarrow$ NMOS: Cutoff, PMOS: Triode
- Region-2: $V_{TN} \leq V_{in} < V_{Ipt} \rightarrow$ NMOS: Saturation, PMOS: Triode
- Region-3: $V_{Ipt} \leq V_{in} < V_{Int} \rightarrow$ NMOS: Saturation, PMOS: Sat.
- Region-4: $V_{Int} \leq V_{in} < (V_{DD} - |V_{tp}|) \rightarrow$ NMOS: Triode, PMOS: Sat.
- Region-5: $(V_{DD} - |V_{tp}|) \leq V_{in} < V_{DD} \rightarrow$ NMOS: Triode, PMOS: Cutoff

Critical voltages on the VTC: V_{opt}, V_{Int}, V_{It} (Considering Ideal VTC)

$$V_{It} = \frac{\sqrt{K_p} (V_{DD} - |V_{tp}|) + \sqrt{K_n} V_{TN}}{\sqrt{K_n} + \sqrt{K_p}}$$

$$V_{opt} = V_{Ipt} + |V_{tp}| = V_{It} + |V_{tp}|$$

$$V_{Int} = V_{Int} - V_{TN} = V_{It} - V_{TN}$$

Derivation of Critical Voltages

V_{It} :

During $V_{It} \rightarrow$ both M_n & M_p are in transition point.

So for $M_p \rightarrow I_{Dp}(sat) = I_{Dp}(linear)$

and for $M_n \rightarrow I_{Dn}(sat) = I_{Dn}(linear)$

again here $I_{Dp} = I_{Dn}$

we take $\rightarrow I_{Dp}(sat) = I_{Dn}(sat)$ for simplicity

$$\Rightarrow K_p (V_{Sop} + V_{tp})^2 = K_n (V_{asn} - V_{tn})^2$$

$$\Rightarrow K_p (V_{DD} - V_{It} + V_{tp})^2 = K_n (V_{It} - V_{tn})^2$$

at this point $\rightarrow V_{It} = V_{It}$

$$\Rightarrow \sqrt{K_p} (V_{DD} - V_{It} + V_{tp}) = \sqrt{K_n} (V_{It} - V_{tn})$$

$$\Rightarrow \sqrt{K_p} (V_{DD} + V_{tp}) - \sqrt{K_p} V_{It} = \sqrt{K_n} V_{It} - \sqrt{K_n} V_{tn}$$

$$\Rightarrow V_{It} (\sqrt{K_p} + \sqrt{K_n}) = \sqrt{K_p} (V_{DD} + V_{tp}) + \sqrt{K_n} V_{tn}$$

$$\therefore V_{It} = \frac{\sqrt{K_p} (V_{DD} + V_{tp}) + \sqrt{K_n} V_{tn}}{\sqrt{K_p} + \sqrt{K_n}}$$

or,

$$\therefore V_{It} = \frac{\sqrt{K_p} (V_{DD} - |V_{tp}|) + \sqrt{K_n} V_{tn}}{\sqrt{K_p} + \sqrt{K_n}}$$

V_{ont} :

M_n is in transition point

$$\therefore V_{DS_n} = V_{GS_n} - V_{tn}$$

$$\Rightarrow V_o = V_{It} - V_{tn}$$

at this point $\rightarrow V_o = V_{ont}$,

$$V_{It} = V_{It}$$

$$\therefore V_{ont} = V_{It} - V_{tn}$$

V_{opt} :

M_p is in transition point

$$\therefore V_{SD_p} = V_{SG_p} - |V_{tp}|$$

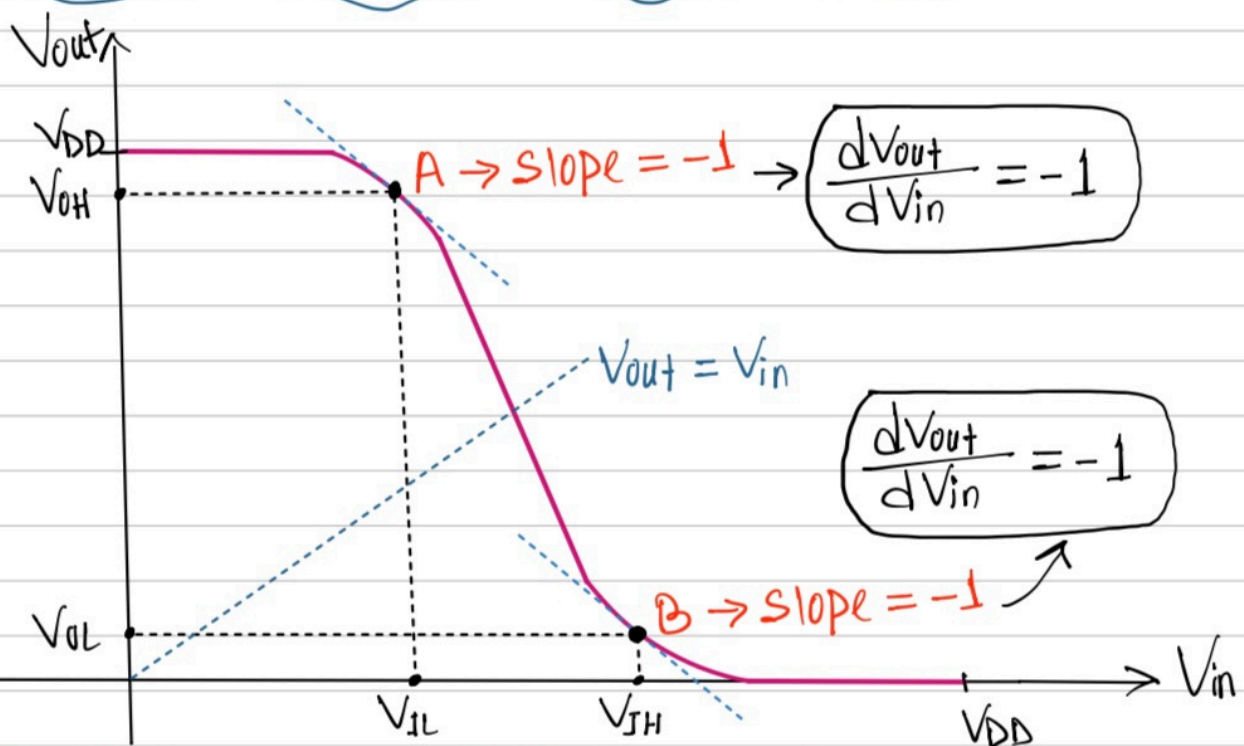
$$\Rightarrow V_{DD} - V_o = V_{DD} - V_{It} - |V_{tp}|$$

at this point $\rightarrow V_o = V_{opt}$,

$$V_{It} = V_{It}$$

$$\therefore V_{opt} = V_{It} + |V_{tp}|$$

Noise Margin of CMOS Inverter



→ Point A and B → inflection point → Between these two points, the slope of the VTC is the steepest. So, rapid changes occurs in output for a small changes in the input voltage. Slope = -1 at points A & B

Low state Noise Margin, $NM_L = V_{IL} - V_{OL}$

High state Noise Margin, $NM_H = V_{OH} - V_{IH}$

At point - A → NMOS: Saturation, PMOS: Triode
 ↳ $I_{ON}(\text{Sat.})$ ↳ $I_{DP}(\text{Triode})$

So, we can determine V_{OH} and V_{IL} by solving $I_{ON} = I_{DP}$
 where $\frac{dV_{OH}}{dV_{IL}} = -1$ (Slope = -1)

At point - B → NMOS: Triode, PMOS: Saturation
 ↳ $I_{ON}(\text{Triode})$ ↳ $I_{DP}(\text{Sat.})$

So, we can determine V_{OL} and V_{IH} by solving $I_{ON} = I_{DP}$
 where $\frac{dV_{OL}}{dV_{IH}} = -1$ (Slope = -1)

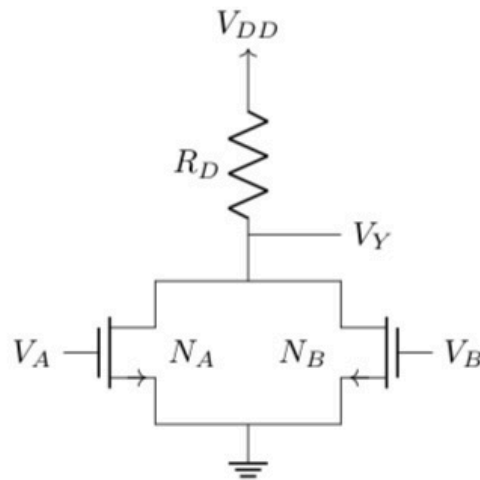
Problems Roster

Problem 1:

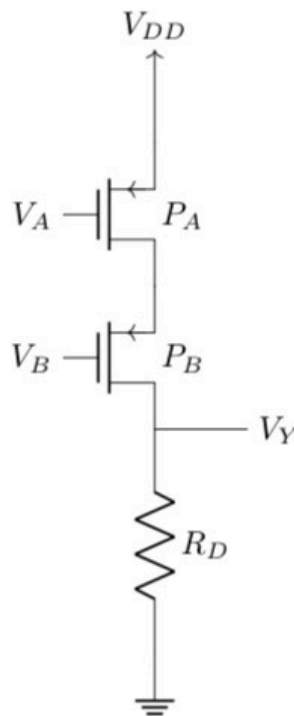
Implement a two-input NOR gate using CMOS logic.

Solution:

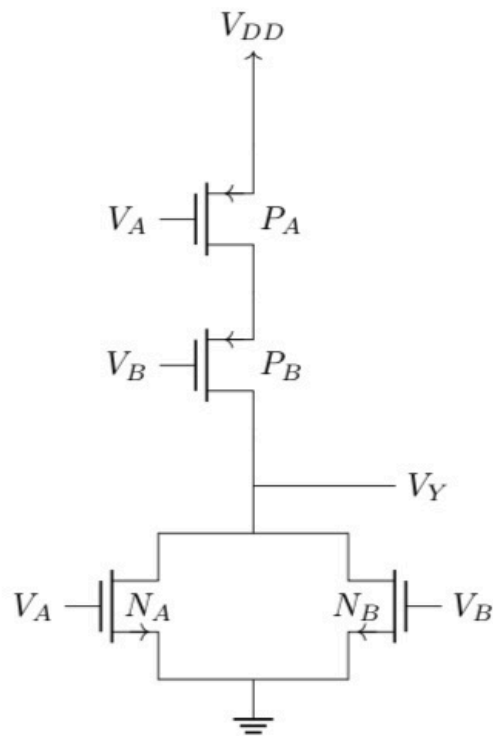
NOR gate using NMOS—



So, the NMOS network has two NMOS in parallel. Thus, according to conduction complement, the equivalent PMOS network will have two PMOS in series. If we had intended to implement the NOR gate using PMOS only, the circuit would be,



To get the CMOS network, we discard the resistances of both the NMOS & the PMOS networks, and connect their respective output nodes together. Thus, the CMOS circuit would be,

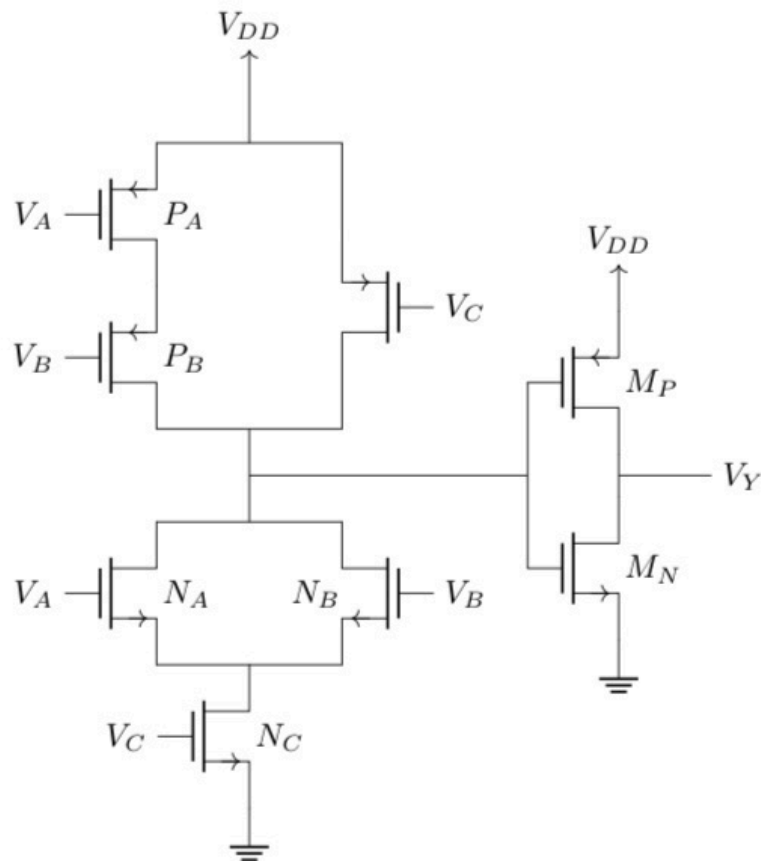


Problem 2:

Implement the logic function $Y = (A+B)C$ using CMOS logic.

Solution:

Since the function has no bar over it, we start with designing $Y' = \overline{(A+B)C}$. Then we will use a CMOS inverter to get the actual 'Y'. So, $Y = \overline{Y'}$.



Problem 3:

Determine the critical voltages on the voltage transfer curve of a CMOS inverter. Consider a CMOS inverter biased at $V_{DD} = 5V$, with $K_N = K_P$ and $|V_{TP}| = V_{TN} = 0.8V$.

Solution:

Using 10.1, we can find V_{It} as,

$$V_{It} = \frac{\sqrt{K_P}(V_{DD} + V_{TP}) + \sqrt{K_N}V_{TN}}{\sqrt{K_N} + \sqrt{K_P}} = \frac{(5 - 0.8) + 0.8}{2} = 2.5V$$

Now,

$$V_{OPt} = V_{It} - V_{TP} = 2.5 + 0.8 = 3.3V$$

And,

$$V_{ONt} = V_{It} - V_{TN} = 2.5 - 0.8 = 1.7V$$

Problem 4:

Determine the noise margin of a CMOS inverter for the given specifications, $V_{DD} = 5V$, with $K_N = K_P = 50 \mu A/V^2$ and $-V_{TP} = V_{TN} = 1V$.

Solution:

Finding V_{IL} & V_{OH} : M_N is in saturation and M_P is in triode. So,

$$\begin{aligned} I_{D_N} = I_{D_P} &\Rightarrow K_N(V_{GS_N} - V_{TN})^2 = K_P[2(V_{SG_P} + V_{TP})V_{SD_P} - V_{SD_P}^2] \\ &\Rightarrow K_N(V_{IL} - V_{TN})^2 = K_P[2(V_{DD} - V_{IL} + V_{TP})(V_{DD} - V_{OH}) - (V_{DD} - V_{OH})^2] \\ &\Rightarrow 50 \times 10^{-6}(V_{IL} - 1)^2 = 50 \times 10^{-6}[2(5 - V_{IL} - 1)(5 - V_{OH}) - (5 - V_{OH})^2] \end{aligned}$$

Differentiating both sides w.r.t. V_{IL} ,

$$2(V_{IL} - 1) = -2(5 - V_{OH}) + 2(4 - V_{IL})\left(-\frac{dV_{OH}}{dV_{IL}}\right) - 2(5 - V_{OH})\left(-\frac{dV_{OH}}{dV_{IL}}\right)$$

Since at the critical point, $\frac{dV_{OH}}{dV_{IL}} = -1$,

$$\begin{aligned} 2(V_{IL} - 1) &= -2(5 - V_{OH}) + 2(4 - V_{IL}) - 2(5 - V_{OH}) \\ &\Rightarrow V_{IL} = V_{OH} - 2.5 \end{aligned}$$

Substituting this in the original equation,

$$(V_{OH} - 2.5 - 1)^2 = [2(5 - (V_{OH} - 2.5) - 1)(5 - V_{OH}) - (5 - V_{OH})^2]$$

Solving this we get, $V_{OH} = 4.625V$ and $V_{IL} = 2.125V$.

Finding V_{IH} & V_{OL} : M_N is in triode and M_P is in saturation. So,

$$\begin{aligned} I_{D_N} = I_{D_P} &\Rightarrow K_N[2(V_{GS_N} - V_{TN})V_{DS_N} - V_{DS_N}^2] = K_P(V_{SG_P} + V_{TP})^2 \\ &\Rightarrow [2(V_{IH} - V_{TN})(V_{OL}) - V_{OL}^2] = (V_{DD} - V_{IH} + V_{TP})^2 \\ &\Rightarrow [2(V_{IH} - 1)(V_{OL}) - V_{OL}^2] = (5 - V_{IH} - 1)^2 \end{aligned}$$

Differentiating both sides w.r.t. V_{IH} ,

$$2V_{OL} + 2(V_{IH} - 1)\frac{dV_{OL}}{dV_{IH}} - 2V_{OL}\frac{dV_{OL}}{dV_{IH}} = -2(4 - V_{IH})$$

Since at the critical point, $\frac{dV_{OL}}{dV_{IH}} = -1$,

$$\begin{aligned} 2V_{OL} - 2(V_{IH} - 1) + 2V_{OL} &= -2(4 - V_{IH}) \\ &\Rightarrow V_{OL} = V_{IH} - 2.5 \end{aligned}$$

Substituting this in the original equation,

$$[2(V_{IH} - 1)(V_{IH} - 2.5) - (V_{IH} - 2.5)^2] = (5 - V_{IH} - 1)^2$$

Solving, $V_{OL} = 0.375V$ and $V_{IH} = 2.875V$.

Thus, the noise margins are,

$$NM_L = V_{IL} - V_{OL} = 2.125 - 0.375 = 1.75V$$

$$NM_H = V_{OH} - V_{IH} = 4.625 - 2.875 = 1.75V$$